THE AESTHETICS OF CAUSALITY

A descriptive account into *Ecological Performativity*: a creative research practice.

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ABSTRACT

This article presents some of the contextual frameworks that have located the author's development into a creative research practice calls Ecological Performativity. This practice has evolved from a number of non-linear audiovisual works that are intrinsically linked to geographic and everyday phenomena. These works explore the relationship of environment, material, and process, and are derived from an intensive data gathering procedure and immersion within the respective environments. The project is situated in an ecological discourse that seeks to explore conditions and methods of for co-compositional processes between human and nonhuman bodies. This article negotiates the relational interplay between first person (my) experience in creative practice with that of the interdisciplinary influences that accompany it. These include a number of recent critical, theoretical, and philosophical discourses occurring in the humanities and social sciences generally referred to as The Nonhuman Turn. It is out of this relational interplay that the notion of Ecological Performativity has evolved.

1. INTRODUCTION

> "As techno-science increasingly reaches into every aspect of life, formerly fast held distinctions between the inert and the active, the human and non-human and life and matter are cracking."
> (Salter 2012: 17)

In April 2011 researchers from the natural and social sciences, the humanities, and a variety of creative practitioners gathered in Rotterdam, The Netherlands. Under the title *The Vibrancy Effect: An Anti-Disciplinary Meeting,* the focus was to discuss and "explore the aesthetic-political-technical-ethical effects of vibrant matter" (Salter 2012: 8). The term *Vibrancy,* here, is in direct reference to Jane Bennett's concept of vibrant materiality or "thing-power" that, as Bennett claims, attempts to give voice to the energetic vitality intrinsic to matter and the active, earthy, and complex entanglements of the human and nonhuman (Bennett 2010: 3).

At this meeting, participants presented their unique understandings, approaches, and concerns for considering vibrant materiality, or, what sociologist of science Andrew Pickering calls "material agency"—"the material that comes at us from outside the human realm" (Pickering 1995: 6). The variety of disciplines, terms, crossovers, and paralleling conditions that each practitioner can be traced (Salter et al. 2012: 14) underpins the telos in which this meeting resonated—one of a transdisciplinary, or rather, "anti-disciplinary"[1] enterprise that focused on matters of human, nonhuman and material agency.

Jump ahead to May 2012 and a gathering of scholars at the University of Wisconsin, Milwaukee, this time under the rubric of *The Nonhuman Turn,* and similar to *The Vibrancy Effect*, the discourses to emerge explored the agency of the human and nonhuman. Bennett suggests the relevance of such a turn is "to find new techniques, in speech and art and mood, to disclose the participation of nonhumans in "our" world" (Bennett 2015: 224–225). As noted in the subsequent book from this conference, *The Nonhuman Turn* evolved from a variety intellectual and theoretical developments that occurred within the last decades of the twentieth century: Actor-network theory (Latour), Affect theory (Massumi), Animal studies (Haraway), Assemblage theory (Deleuze), New Materialism (Bennett), Speculative realism (Harman). Broadly speaking, the nonhuman turn can refer to objects such as "climate change, drought, and famine; to biotechnology, intellectual property, and privacy; to genocide, terrorism, and war" (Grusin 2015: vii). Such wide-ranging perspectives on what constitutes an object and nonhuman are, as Salter suggests above, a disruption of distinctions. But given the many concerns arising in the twenty-first century the turn towards the nonhuman has particular relevance to, as Timothy Morton suggests, "exit modernity—which the current ecological emergency seems to be demanding" (Morton 2013b: 80).

2. ISSUES OF AGENCY

> "Thinking issues of agency through the experiential encounter with the 'stuff of the world' encourage a radically different vision of the world—dynamic, temporally emergent, contingent, and performative."
> (Salter 2015)

Thinking in terms of agency and performativity is nothing inordinately new, and in Western thought has evolved from a variety of philosophical, scientific and artistic research that took place over the last century (Salter 2010, 2015; Capra 2014). Of late, however, there

[1] Pickering defines the undertaking of this meeting as a nonmodern ontology of "anti-disciplinarity" (2013: 209–220).

has been a reinvestigation into the notion of agency and performativity that, as Chris Salter suggests above…"is encouraging a radically different vision of the world." From Karen Barad's "intra-action" (2007) and Andrew Pickering's "dance of agency" (1995, 2008) to Jane Bennet's "thing-power" (2010), Tim Ingold's "mesh-work" (2007, 2011) and Timothy Morton's "hyperob-ject" (2013a) a reconceptualization is taking place which challenges the long-standing narrative of human exceptionalism and our understanding of the geo-conjunctures that make up life on earth.

As these thinkers grapple with the notion of agency in human and nonhuman bodies, a host of ecological, social, cultural, and political observations and concerns are being raised and challenged. The urgency of which is energized by what has now been embraced as the Anthropocene;[2] the epoch in which the effects of fossil-fuel-burning humans have fundamentally altered the earth's geological composition.

From the position of creative-research, the attentive-ness to these emerging discourses provides an opportunity to ask new questions of the *making-doing-thinking*[3] of artistic practice. I was specifically drawn to these discourses as a means to contextualize my own creative questions and concerns: What does making art from these lived and experiential encounters in the world do? In other words, what is the purpose of an art form relationally situated in time and place? Since the 1990s my creative practice has revolved around the exploration of the day-to-day situated encounters in the real world (Figure 1). These works were deeply embedded in time and place and explored the impact that human activities have had on the environment. Since then, this practice has evolved from a fixed-media format to one that explores non-linear systems using components such as weather, meteorological, and environmental data. My field recordings have taken place throughout North America, New Zealand, and Australia resulting in a catalogue of audiovisual works that are intrinsically linked to geographical factors and everyday phenomena.



Figure 1: The work *Motion Parallax* (1998) was the first large-scale audiovisual collaboration between Andrew Denton and I and was created using field recordings captured on a cross-Canada trip from Tofino, British Columbia to Cape Spear, New-foundland. This photo was taken on the last day of shooting on top of Signal Hill historical site located in St. John's NFLD.

But what is it about these experiential encounters with the "stuff-of-the-world" that has held my curiosity? What significance does it have on my mode of artist practice, and how does this practice motivate the conditions in which creative possibilities are activated, assembled, and processed? More specifically, what would motivate my collaborators and I to venture on field recordings that would place us in Death Valley at 53 degrees Celcius, the polluted wastelands of the Salton Sea in Southern California, the crowded sidewalks of Los Angeles, and the tourist-filled paved pedestrian trails in Sequoia National Park? The answer to these questions, I believe, resides in practice.

By reorienting my creative practice with these different "modes of thinking,"[4] the process that I refer to as *Ecological Performativity* has evolved. Central to this idea are the fundamental questions: What tendencies emerge in the *making-doing-thinking* of creative practice when material agency is considered a co-compositional device? What capasities do these tendencies have on the creative process and how do they affect the resulting artefact? Might a broadened understanding of agency and performativity provide different vocabularies and networks of communication? Can this encourage an attunement to the reality of the coexistence of all things on Earth? And if so, as a creative practitioner, what, then, is my response and response-ability?

3. ECOLOGICAL PERFORMATIVITY

> "The world is an open process of mattering through which mattering itself acquires meaning and form through the realization of different agential possibilities"
> (Barad 2007)

Open processes and different agential possibilities are central to the creative practice of *Ecological Performativity*. *Ecological* is located within the philosophical provocations of Brian Massumi and Erin Manning as being that of a relational experience: "Organisms-that-person agitate in the mix, but always in a witness of environment: a becoming ecology of practices" (Manning et al. 2014). Thus, this practice considers emergence and material agency as co-creative apparatuses. Accordingly, *Performativity* draws specifi-cally upon Andrew Pickering's notion of the "dance of agency" (1995, 2013). Here, agency and performativity are entwined in what Pickering posits as the performative idiom (2012). This is Pickering's attempt

[2] Ecologist Eugene Stoermer and atmospheric chemist Paul Crutzen first proposed the term Anthropocene as a means to name the geologi-cal transformations that have occurred in the oceans (acidification), coral reefs (bleaching), and raising global temperatures due to the burning of fossil fuels. (Crutzen, et al. 2000)

[3] Haraway describes this figure of practice as a back and forth passing of patterns, similar to string figure games. (Haraway 2014)

[4] I borrow this turn of phrase from Manning and Massumi in which they claim: "Every practice is a mode of thought, already in the act. To dance: a thinking in movement. To paint: a thinking through color. To perceive in the everyday: a thinking of the world's varied ways of affording itself." Each is a technique, or, springboard that sets in mo-tions "a practice from within" (Manning et al. 2014: Kindle edition Loc 52).

to move away from the idea that agency is specific only to humans, or to what he refers to as "human exceptionalism" (2012). He suggests that the world, in all its heterogeneous multiplicity, is full of agency and processes of emergence. By exploring these processes and performative relationships between things, including those beyond the human realm, Pickering suggests that we invite the "possibility of a non-modern stance of revealing rather than enframing which, in turn, invites open-ended extensions" (Pickering 2010).

Similar to other ecologically-grounded creative practices, (Barclay 2013; Burtner 2011; Di Scipio 2011; Keller et al. 2006; Opie et al. 2006) *Ecological Performativity* explores the relationships of environment, material, and process, and is derived from an intensive data-gathering procedure and immersion within the respective environments. Each work begins in a collaborative field recording process that often starts in a matter-of-fact manner (making sure all batteries are charged etc.). However, the effect these environments have on my collaborators and I become an operative agent—attunement of which takes time. According to Bennett it requires: "a cultivated, patient, sensory attentiveness to nonhuman forces operating outside and inside the human body" (2010: xiv). The agency of such forces is relational. Bennett's discourse on "thing-power" surmises that: "Earthy bodies, of various but always finite durations, affect and are affected by one another. And they form noisy systems or temporary working assemblages that are, as much as any individuated thing, loci of effectivity and allure" (2015: 233). Of this my long time collaborator Andrew Denton writes: "Once time is taken to absorb [the location], I attempt to record material that communicates my sensations and experiences of being there" (2016). He reflects that by "letting go of a need to understand, comprehend, and categorize […] the intensity of the making-feeling-thinking [could] take over in the moment of capture, leave[ing] the reflection and reinterpretation for a later distanced encounter with the material during post-production" (2016).

There is a causal dimension that, as Morton argues, is "wholly an *aesthetic* phenomenon" (Morton 2013b). Accordingly, Morton proposes a wide-ranging account of aesthetical moments. "Aesthetic events are not limited to interactions between humans or between humans and painted canvases or between humans and sentences in dramas. They happen when a saw bites into a fresh piece of plywood. They happen when a worm oozes out of some wet soil. They happen when a massive object emits gravity waves" (Morton 2013b: 19).

3.1 The Ecology of Practice

The post-production exploration of materials is done in part through the development of specifically designed computational systems. These systems vary in construction and are intrinsically linked to the collected location data of audio field recordings, moving images and photos, as well as weather, meteorological, and environmental data gleaned from these situated encounters. Through research the techniques include computer vision processes, data sonification, live convolution, and improvisation as a mean to engage the agency of material and thus construct the non-linear audiovisual installations. What emerges does so in an iterative manner that affords an open-ended interaction in the "ecology of practice." Similar to Massumi and Manning, Isabelle Stengers says:

> "[A]n ecology of practice is a tool for thinking through what is happening, and a tool is never neutral. A tool can be passed from hand to hand, but each time the gesture of taking it in hand will be a particular one—the tool is not a general means, defined as adequate for a set of particular aims, potentially including the one of the person who is taking it, and it does not entail a judgement on the situation as justifying its use. […] Here the gesture of taking in hand is not justified by, but both producing and produced by, the relationship of relevance between the situation and the tool" (2005: 185).

Approaching my research accordingly emphasizes the importance of a mode of practice that is situational, emergent, and able to diverge in the process of experimentation. I consider these three components tools and techniques in that the *relationship of relevance* (Stengers) draws forward the thinking-feeling that can traverse all movements of experience. Technique, as understood by Manning and Massumi, "belongs to the act" and are springboards that sets in motion "a practice from within" (2014: Loc 71) Here, then, the subjective and objective are not positioned on opposite planes but rather move in a relational field that mobilises and transforms the work. Mannning and Massumi reflect: "Thought gathers in the work. It is the event of the work's unfolding. Not into language, but painting, on a canvas that seeks to activate a new way of seeing, a new effort at participation" (Ibid). In my research, the canvas equates to the resulting non-linear audiovisual installations while the effort to participate is experienced in the field and in the act of *making-doing-thinking*.

Subsequently, this ecology of practice has come to involve the recording of live musical improvisations in response to the developed system. This has become an important component of *Ecological Performativity,*— which is within the iterative developments of these systems out of the material gathered; an acoustic musician is then invited into the process to respond improvisationally to the material. Recordings have taken place in live multimedia concert improvisations, studio settings, and the respective environments. What this provides is a cumulative database that in turn folds back into the final system. Motivated by the desire to explore non-linear systems, the installation platform provides a space where the constraints of beginnings, middles, and ends are eliminated. The artwork can then exist as a transformative apparatus.

Operating in this discursive register, from the core of creative research, provides a platform for experimentation-as-process that contributes to new ways of thinking by insisting that every practice is a knowledge that can speak and act through the differences and emerging possibilities.

4. Dark Ecology, the Sonic Potentials of Data and the Salton Sea

> "With dark ecology, we can explore all kinds of art forms as ecological: not just ones that are about lions and mountains […]. The ecological thought includes negativity and irony, ugliness and horror" (Morton 2010: 17).

Anybody who has ventured into the writings of Timothy Morton will be familiar with the complexity of ideas spun on every page. From his book *Ecology Without Nature* and ideas of the "hyperobject" to his dark ecological thoughts, Morton's philosophical ponderings purpose a way of thinking and being (of which he considers thinking, in and of itself, an ecological event) that embraces ambiguity, uncertainty and the uncanniness of the entangled mesh. Morton is a strong advocate for art, philosophy, and music stating that: "…art forms have something to tell us about the environment, because they can make us question reality" (Morton 2010: 8).

When considering Morton's idea of the hyperobject, that of, "agents or objects so massively distributed in time and space as to transcend localization, such as the biosphere, global warming, or the sum of all the whirring machinery of capitalism," (Morton 2013a: Loc 110) the creative practice of making works form field recordings and data becomes multifaceted. When one reflects on the interwoven interactions that occur in any given encounter; between what is seen and unseen, heard and inaudible to our human ears, the complexity of the mesh is immense. For Morton, "the mesh" substitutes words such as interdependence and interconnectedness (2010: 28). For Tim Ingold, the mesh is a metaphor for the relational interwoven lines of lived experience (2007: 103). Borrowing from philosopher Henri Lefebvre, Timothy Ingold defines meshwork as a "zone of entanglement [where] there are no insides or outsides, only openings and ways through" (Ibid). In my creative research, thinking in terms of the mesh underpins the practice of *Ecological Performativity*. By engaging in a non-deterministic way with what is present in any given environment, "the poetic potential of locational data has the capacity to draw you to the multiplicity and complexity of the content" (Denton 2016).

This practice was put into play when collaborators Andrew Denton, Adrian McNaught and I recently embarked on an audiovisual collection process throughout the Southwestern drought regions of the United States. This three-week field recording session involved many extreme locations including Bombay Beach on the Salton Sea. Parking our vehicle and venturing into this environment, the odour itself stopped us in our tracks. The shoreline was littered with dead fish and birds and human objects in varying stages of decline, all of which were covered with a dusty white mixture of salt and dried thermal mud. This environment is the result of early 20th century weather systems and ensuing human activities.

In 1905, when the Colorado River swelled and breached its banks, the water ran into the Salton Sink, a geographical region 220 feet below sea level. After two years of continuous flow, a 15-by-35 mile lake formed that became known as the Salton Sea.[5] Taking advantage of California's newest and now largest lake, the Salton Sea became a favorite getaway spot for nearby Los Angeles and San Diego residents. During the 1950s and '60s, Bombay Beach, which is located on the lake's eastern side, became a prosperous resort town filled with sunbathers, water-skiers, and yacht club parties. During the 1970s, however, it became apparent that the ecosystem of the Salton Sea was quickly deteriorating. With no drainage outlet and little to no annual rainfall, the inflow of industrial pollutants and untreated sewage began to increase the lake's salient level and caused the water to deoxygenate. What had become an angler's well-stocked paradise quickly transformed into a rotten layer of dead fish and birds (Paiva).

The indexical signs of the human and nonhuman now litter Bombay Beach, which has been described as "the most depressing place in California" (Riggs 2010). Once Denton, McNaught, and I had adjusted to the initial shock of this environment, we proceeded to record these indexical signs. Denton and McNaught focused on the visual components, while I focused on the sonic environment. I became transfixed with the numerous objects scattered throughout: rusty metal objects sticking out of the ground, wooden refuse from dilapidated buildings, sections of concrete slab, plastic bags entangled and flapping in dead bushes, and a lone broken piano (Figure 2). Using contact microphones, I recorded the sonic textures and tones by tapping, plucking and playing these objects. Equally striking was the sound resounding at the waters edge. Primarily comprised of crushed fish and bird bones, the sonic quality activated by wave and human footsteps has a sharp percussive high-pitched resonance. I captured this using a hydrophone.

It was during this field recording I posed the question to my collaborators: what is the purpose of making art from these lived and experiential encounters? Beyond technical and aesthetic choices, the creative research nexus attends to the frailty, vulnerability and performative substance of time and place. Morton surmises "to be located "in" space or "in" time is already to have been caught in a web of relations" (Morton 2013b: 21). From



Figure 2: Piano on Bombay Beach. Photo Adrian McNaught

[5] http://www.sci.sdsu.edu/salton/Salton%20Sea%20Description.html



Figure 3 Birds in flight on the Salton Sea. Photo Andrew Denton

a sonic arts practice Kim-Cohen suggests that: "Every work of art is a response to the conditions within which it is produced and received […], the assumptions and problems inherent to its time and place" (Kim-Cohen 2013: Loc 100). Or, perhaps, by choosing to engage with the negativity, irony and ugliness of these environments—Morton's dark ecology, the capacity to recalibrate the world through our practice is opened by drawing out the evocative and emotional that, in turn, provides the opportunity to see, hear and be *in* the world differently (Denton 2016: 64). The artwork becomes an apparatus of change.

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